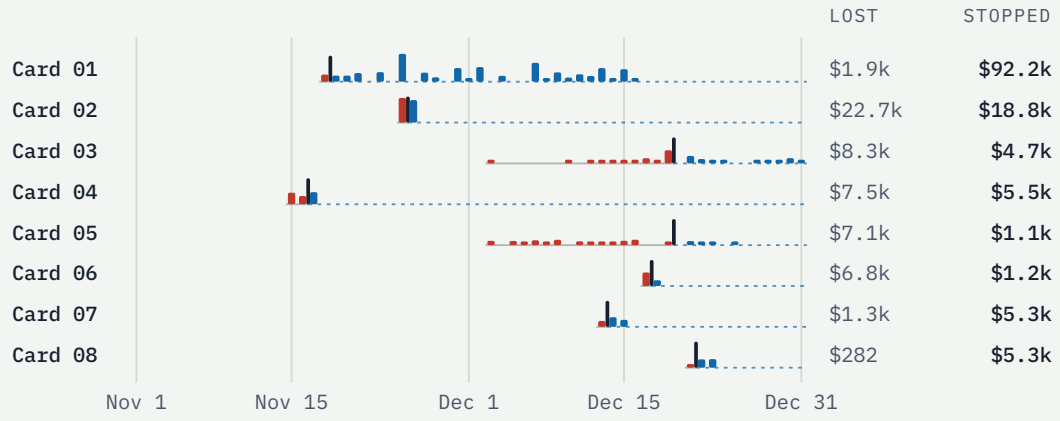
Stopping the card, not the transaction

Should the program deploy a fraud score, and how fast should it act on an alert?

Jay Yu · UCLA Anderson, MS in Business Analytics



Score every purchase and block the card the same day an alert is confirmed: the model stops 81% of fraud dollars for under a tenth of an analyst.

64.5%

of fraud transactions land in the top 3% of scores on months the model never saw (Nov–Dec).

$170k of $209k

of fraud stopped with same-day blocking; next-day blocking stops 69%.

1.6 alerts/day

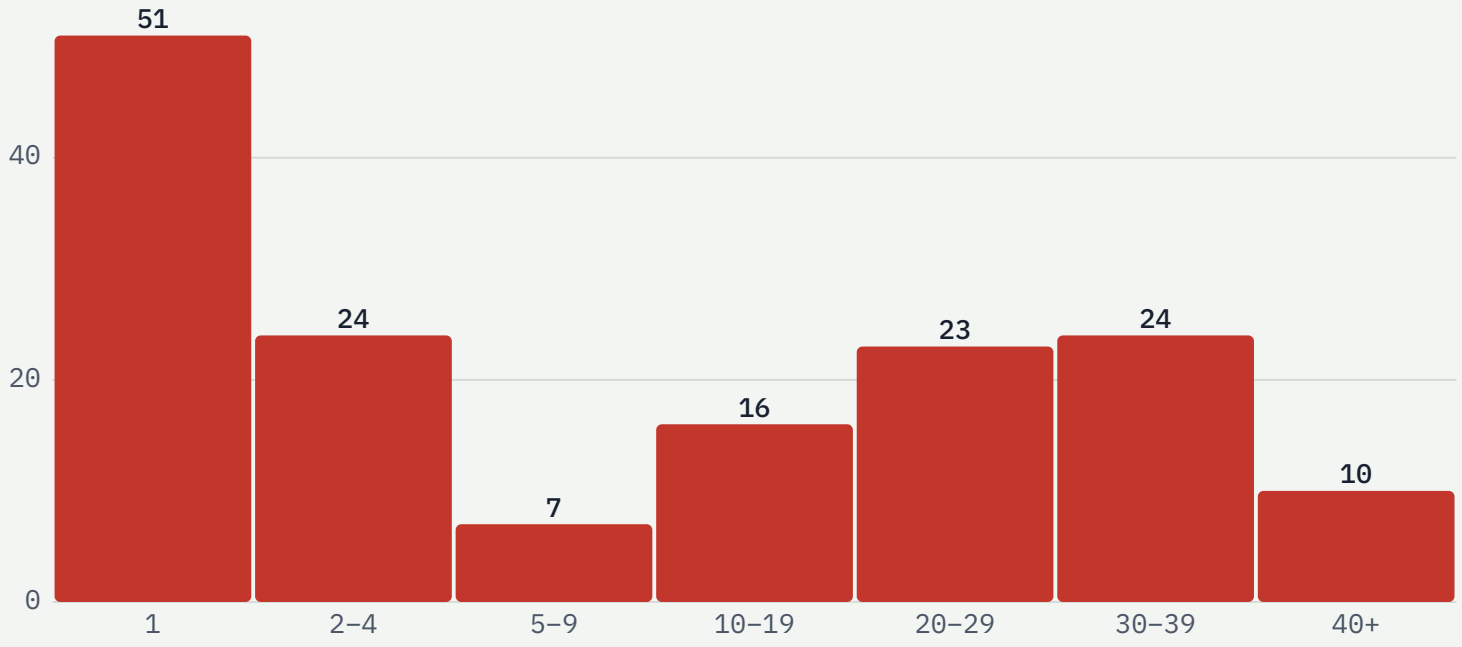
at this program's volume, about 16 minutes of analyst time a day.

DECISION REQUESTED

Approve a 60-day shadow pilot: score live purchases, work alerts the same day, and go live if the top 3% of scores catch at least 45% of fraud.

Fraud is concentrated by card and front-loaded: 39% of fraud dollars land on a card's first fraud day.

Compromised cards by number of fraud purchases, 2010



155

cards carry all 2,283 fraud purchases, out of 1,643 cards.

58%

of fraud dollars sit on the ten costliest cards.

2.3%

of purchases are fraud: rare per purchase, costly per card.

97,805 purchases, Jan–Dec 2010, after exclusions

Five steps from raw purchases to a staffing decision, every one judged on later months.

1

Clean

Purchases only; merchant, state and ZIP recovered without touching the label.

97,805

purchases kept

2

Describe behavior

Point-in-time counts, spend and bursts per card, merchant and pairing, 0–30 days back.

224

variables

3

Model on time order

Boosted trees tuned on two forward-in-time folds; Nov–Dec never used to choose.

49

settings compared

4

Replay card blocks

Nov–Dec purchase by purchase: alert, confirm, block, and count what the block stops.

16

compromised cards

5

Price the decision

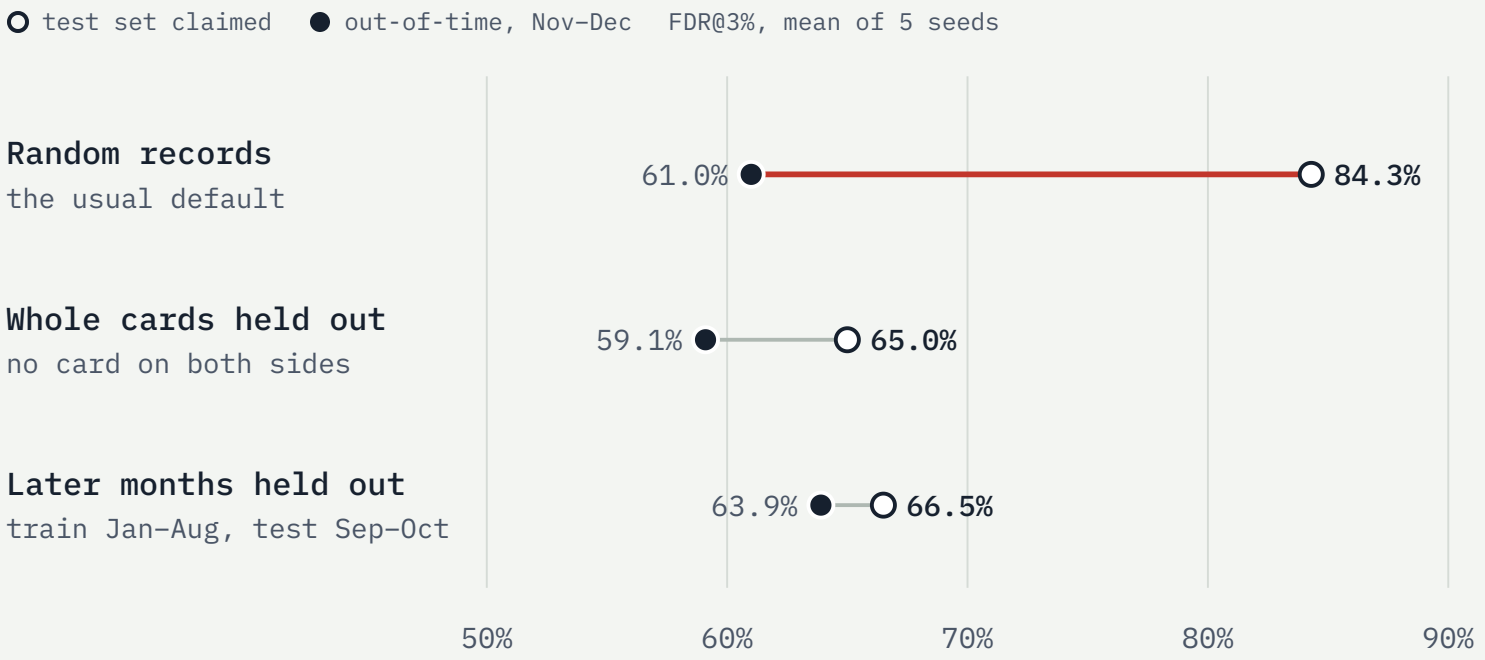
Net value by review budget, analyst capacity and response speed, with every cost varied.

5,000

simulated outcomes

Everything is one reproducible pipeline: `fraudscore run → tune → decide → export`, with unit tests for every rule that could leak the future into the past.

A random test split promised 84% detection. On later months the same model found 61%, so we tune and report on time order only.



96%

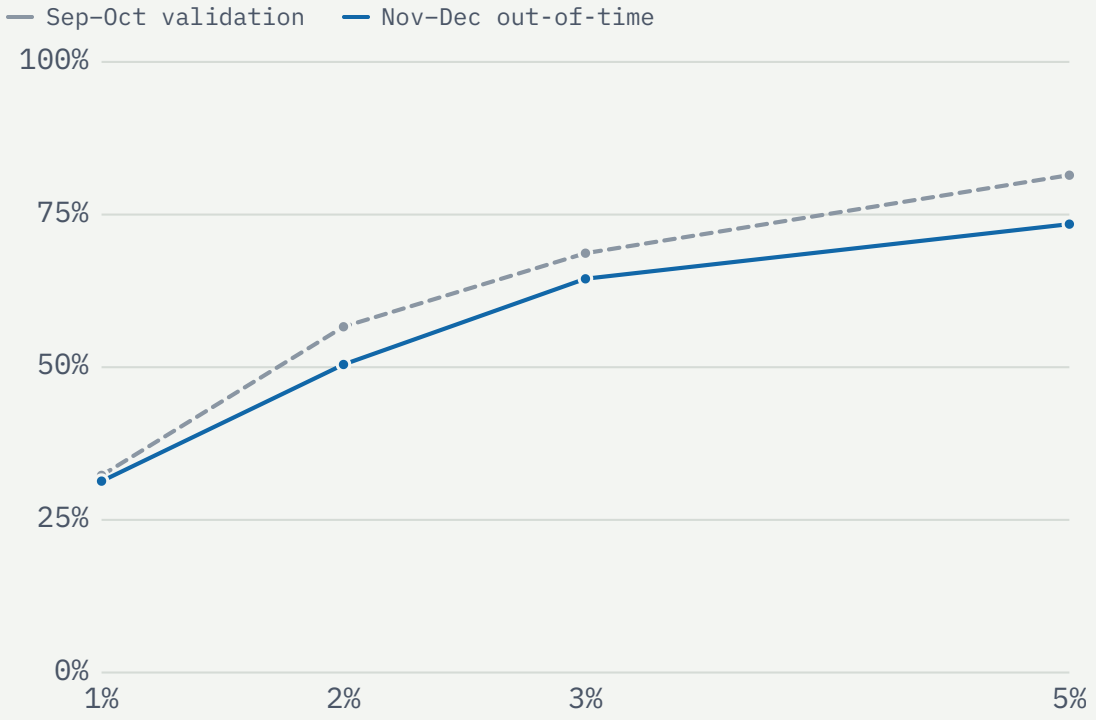
of the random split's test frauds sat on a card whose other frauds were in training. The model recognized the card, not the pattern.

What changed

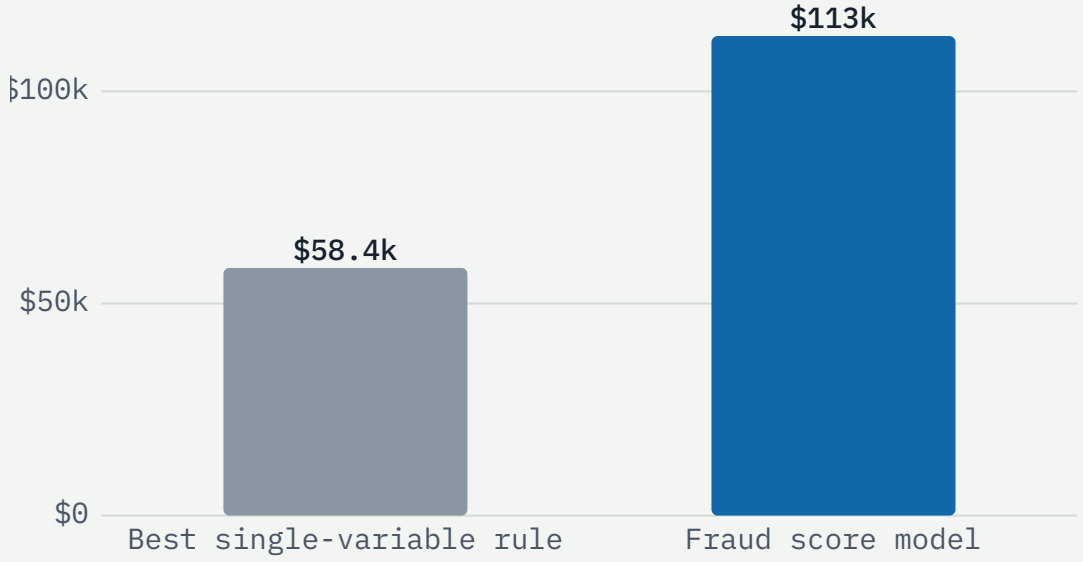
Settings were chosen on two forward-in-time folds; Nov–Dec was held out from every choice, including the alert cutoff.

The top 3% of scores hold 64.5% of Nov–Dec fraud, worth about twice the best rule an analyst could write.

Share of fraud found, by review budget



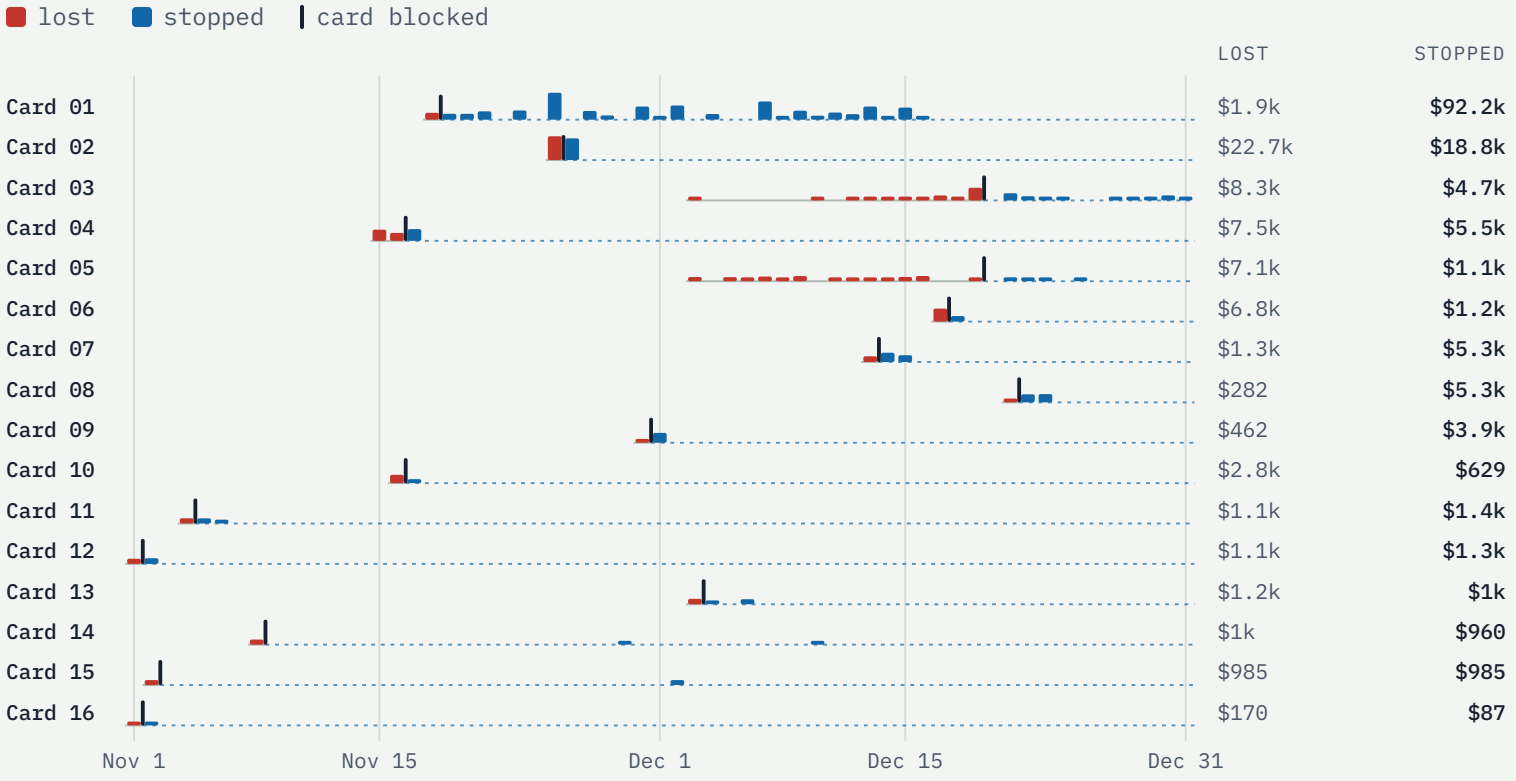
Net value over Nov–Dec, next-day blocking



The rule is a card's spend over the last 3 days, the strongest single variable, tuned on the same months.

Speed is the lever: with the same alerts, blocking the same day stops 81% of fraud dollars instead of 69%.

Every compromised card, Nov 1 – Dec 31, next-day blocking



69%

stopped when the card is blocked the next morning.

81%

when alerts are worked and cards blocked the same day: $25.6k more, at the cost of 16 more good purchases declined.

88%

with real-time declines, but 143 good purchases declined. A later step, not the first.

Same model and cutoff for every policy · good purchases declined include the 5-day reissue lag

Start with a 60-day shadow pilot whose go-live bar matches what 16 compromised cards can prove.

DAYS 1–60

Shadow

Score every purchase, work the top 3% the same day, log what a block would have stopped. Go live if the top 3% catches ≥ 45% of fraud.

FROM DAY 61

Same-day blocking

Block confirmed cards; keep 10% of cards scored but not blocked as the comparison group.

ONCE FALSE DECLINES ARE KNOWN

Real-time declines

Decline at the point of sale only for the highest scores.

Small samples

A 60-day pilot pins detection to ±16 points. It can confirm the score works, not fine-tune it.

Value range

$24k–$240k net over two months (90%), never negative; driven by which cards get compromised.

Drift

Score stability is 0.048 PSI today. Review above 0.10, retrain above 0.25 or quarterly.